

Aerodynamic Analysis of a High-Camber , Single-Element Rear Wing for Low-Speed Automotive Applications

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I. Abstract –

This paper presents the numerical results of a custom high-camber single-element wing , designed for providing downforce to Formula Student type vehicles at low-speed circuits. The wing was designed with a conservative chord length of 100 mm , due to an absence of the true dimensions of the entire car. For this project , absence of wingtip vortices is assumed ,as the purpose is to purely test the flow separation and downforce generated by this custom designed wing. The wing has a high camber of 4% of chord length and a thickness of 12% of chord length. For this specific project , the sketch was designed directly in a cloud Computer Aided Design (CAD), OnShape , instead of traditional software like XFOIL or XFLR5. Tools such as splines , conics, circles and constraints were used to manually hand-draw the 2D sketch, which was then extruded to finally form the 3D wing. After modelling the wing , it was analysed in a cloud Computational Fluid Dynamics (CFD) software , SimScale. Free-stream velocity of 15 m/s was used during external flow simulation of the wing , using incompressible air flow. Results demonstrate fully attached airflow with a Reynolds number (Re) of approx. 1.10×10^5 . Quantitative force tracking track graph converged to yield a net vertical downforce of 1.91 N and a drag penalty of 0.27 N. This correlates with an isolated lifting efficiency of 7.07 on the centreline plane. This project lays down the framework for future wing analysis of computer-generated aerofoil designs, multi element wing designs and wake optimization.

II. INTRODUCTION AND AERODYNAMIC INTENT

- a) **Vehicle Dynamics & Packaging Constraints :** Since the purpose of the wing was to be used along with a Formula Student car , the dimensions of the wing were constrained by the Formula Student Germany (FSG) rulebook and the physical reality of the speed of Formula Student cars. According to the rulebook , the wingspan must not exceed 1200 mm , which is the max width of the car including

the wheels. For this project a conservative value of 500 mm is assumed as the wingspan, to ensure easy packaging of the wing (if developed later) and prevent structural errors. Thus, to maintain a wing aspect ratio of 5, chord length of the 100 mm is taken.

$$AspectRatio = \frac{b^2}{S} = \frac{b}{c}$$

(b = wingspan, c = chord length)

Furthermore, the wing takes an aggressive profile with high camber and high thickness. This is because Formula Student cars operate at low speed circuits where taking corners efficiently has more priority than raw top speed and low drag penalty at straights. As a result of this, the average speed of these cars are quite low, at around 10-20 m/s. Since aerodynamic forces are proportional to square of velocity, generating high downforce at such low speeds require aggressive wing profiles.

- b) **Aerofoil Profile Selection & Geometric Intent:** Camber is defined as the intentional asymmetry between the top and bottom surfaces of an aerofoil. The camber line shifts the curvature of the wing downward. As a result of this, winds at the lower curve of the wing are forced to accelerate faster than the winds at the upper surface of the wing. This difference of speed between the upper and lower surface of the wing causes a net force to act downwards on the wing, thus producing downforce. To maximize this downforce, a high camber of 4% is used. Thickness of the aerofoil determines structural rigidity of the wing and steepness of the curves. Too high thickness would cause loss of kinetic energy of oncoming winds. Oncoming winds require kinetic energy to overcome the adverse pressure gradient which they face while travelling to the trailing edge of the wing (the gradient is caused due to the difference between atmospheric pressure and the pressure of the oncoming winds). If they do not have enough kinetic energy to overcome the adverse pressure gradient, they will start recirculating and create a wake behind the wing. Flow separation is never ideal as it causes huge drag and the wing does not provide the required downforce. Too low of a thickness will prevent flow separation but would cause loss of downforce of the wing and structural strength.

III. METHODOLOGY & COMPUTATIONAL SETUP

- a) **CAD Modelling & Data Pipeline Constraints :** Before starting the project, the workflow was designed to first use CAD to draw the 2D sketch and then refine it in XFLR5, a software used to optimize 2D aerofoil sketches. The design of the workflow

however was flawed. The CAD software used to make the 2D sketch, OnShape could only export the 2D sketch as .dxf file type (stands for Data Exchange Format), however XFLR5 used only .dat file type. The conversion of the file from .dxf to .dat was very difficult as the converter had to convert mathematically complex equations of curves into a simple list of coordinates, thus there was a bottleneck during file conversion, where the converter had difficulties and threw an error as there were more than two splines used in the sketch. Thus, as a compromise for a missing step in the workflow, the sketch was manually optimized again in the CAD software by smoothening out the curves, before it was sent to a CFD software for results and its analysis. Though there was a lot of optimization potential in this wing, the results are certainly noteworthy and will serve as a baseline for future projects. To ensure smooth curves, splines and conics were used along with the “Equal Curvature” constraint in OnShape. Furthermore, tangents along with “Coincident” and “Parallel” constraints were used to verify the locations of maximum camber and thickness and also to create a smooth transition between the leading-edge circle and the spline. The 2D sketch was then extruded by 500 mm to create the wing. It was then exported as a STEP file to ensure perfect surface curvature.

- b) **Fluid Domain, Boundary Conditions & Results Tracking** : To run the external flow simulation of air along the wing, an external flow volume was created after which the wing was subtracted from the flow volume using a binary operation. To ensure correct results, the velocity inlet was 400 mm in front of the wing to ensure proper convergence of speed of the oncoming wind, and the pressure outlet was 1500 mm behind the wing, to provide enough space for the wake to form (if any). The simulation type was incompressible fluid flow as the velocities did not reach supersonic levels of speed, and the material of fluid used was air of density 1.196 kg/m^3 and kinematic viscosity of $1.529 \times 10^{-5} \text{ m}^2/\text{s}$. The face in front of the wing was assigned the condition of velocity inlet with wind velocity of 15 m/s, while the face behind the wing was assigned the condition of pressure outlet with a gauge pressure of 0 Pa. The remaining 4 walls were all assigned a Slip-wall condition to prevent interaction of the air with the walls. To measure results, a forces and moments plot was added to the conditions. To measure rotational force on the wing (if any), the Centre of Rotation of the wing was assigned to the point exactly $\frac{1}{4}$ th of the chord according to the quarter chord rule.

III. RESULT & AERODYNAMIC PERFORMANCE

a) Global Force Metrics & Aerodynamic Efficiency :

- i. No. of iterations : 1000
- ii. Mesh count : 474.8k – 712.2k

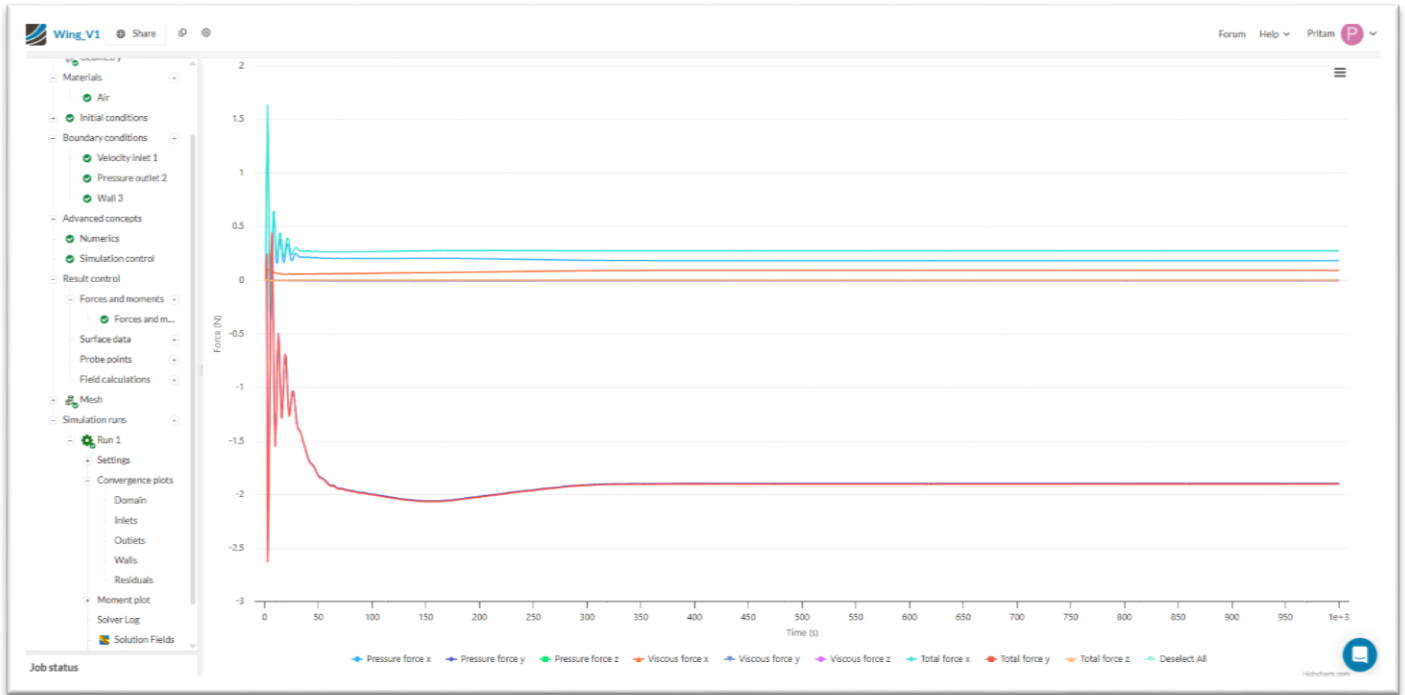


Fig. 1 : Force convergence plot mapping against translational forced in Newtons (N)

Net vertical downforce converged to (F_y) : **-1.91 N**

Net horizontal resistance converged to (F_x) : **+0.27 N**

Calculation of Aerodynamic efficiency :

$$\text{Aerodynamic Efficiency} \left(\frac{L}{D} \right) = \frac{|F_y|}{F_x} = \frac{1.91}{0.27} \approx 7.07$$

Analysis : Note that an efficiency ratio of 7.07 proves the cross-sectional profile successfully generates massive vertical momentum transfer relative to its viscous and pressure drag profile

b) Localized Flow & Field Analysis

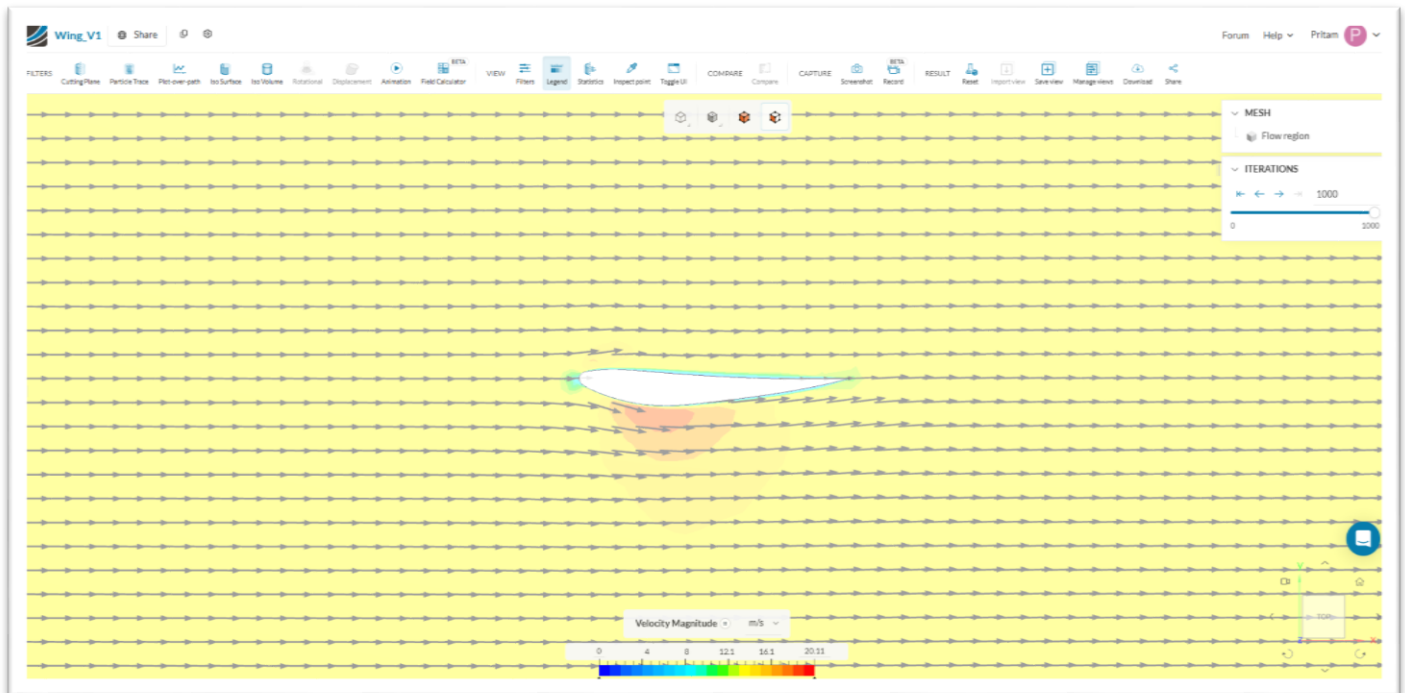


Fig 2: Velocity magnitude profile and vector streamlines along the centreline cutting plane

Analysis : a localized high velocity region , marked by the orange/red colour below the aerofoil , indicates localized acceleration. The difference in velocities between the upper and lower sections of the aerofoil causes downforce according to Bernoulli's principle.

Boundary Layer State : There is a uniform flow of air around the aerofoil as indicated by the green layer. This layer completely tracks the shape of the custom aerofoil without breaking away into a chaotic blue stagnation zone. This indicates that there was no case of flow separation and the flow of air remained attached to the body of the aerofoil throughout its flow.

V. DISCUSSION & FUTURE SCOPE

a) **Comparative Analysis:** An efficiency of 7.07 obtained during this project is not realistic as wingtip vortices were not simulated. Wingtip vortices causes massive drag which results in drop of efficiency of the wing. Furthermore , real race cars often operate under chaotic “dirty air” induced by cars in front.

b) **Future Scope :** This project serves as a base-line for future aerofoil related projects , involving 2D sketch optimization software such as XFOIL or XFLR5. It also serves as a foundation for more complicated simulations involving wingtip vortices and end-plates to counter such vortices.

VI. REFERENCES

- 1) SimScale documentation <https://www.simscale.com/docs/>
- 2) Concepts regarding boundary layer and flow separation
<https://www.youtube.com/@TheEfficientEngineer>

